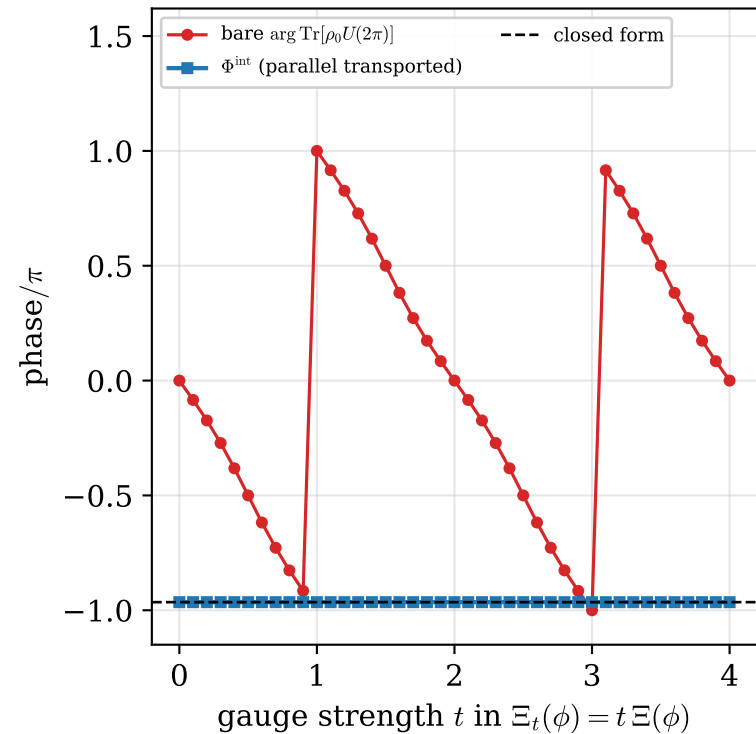
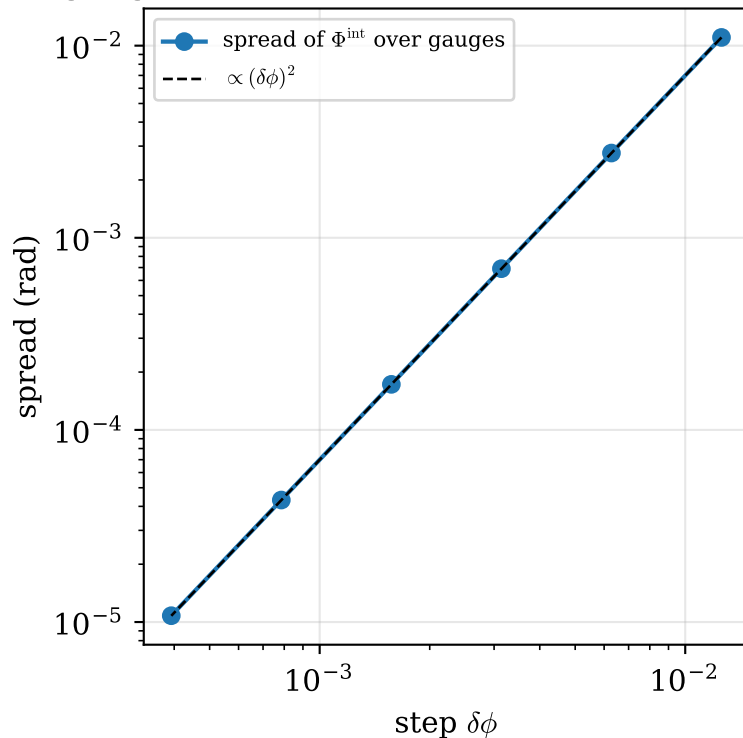


Gauge invariance of the interferometric phase and its single singular point

(a) $U \rightarrow Ue^{i\Xi}$, $[\Xi, \rho_0] = 0$, $\Xi(0) = 0$:
only Φ^{int} is invariant



(b) the residual is pure discretization:
gauge invariance is exact in the continuum



(c) at $C=1$ ($\rho_s = I/2$) the gauge group grows
to $U(2)$: Φ^{int} becomes basis dependent

